

A Simulation of the BB84 Quantum Key Distribution using Qiskit

Abstract

This paper represents a simulation of the BB84 Quantum Key Distribution (QKD) protocol using Qiskit and python. The simulation models secure key exchange through probabilistic quantum measurement and iterative basis alignment. It verifies the core QKD principles by generating, comparing, and filtering quantum bits (qubits) based on basis agreement. A final shared key is used to perform classical encryption and decryption to demonstrate the practical utility of the QKD process.

Introduction

Quantum Key Distribution (QKD) is a secure method for key exchange that leverages quantum mechanics to detect eavesdropping. BB84, proposed by Bennett and Brassard in 1984, was the first QKD protocol. This paper describes the implementation of BB84 using Qiskit. The simulation demonstrates key principles: random bit and basis generation, quantum state preparation and measurement, basis matching, sifting, and shared key agreement.

Mathematical Foundation

The BB84 protocol relies on principles from quantum mechanics, particularly the concept of superposition and measurement collapse.

Two bases are used to encode qubits:

- Z-basis (Computational Basis):
 - Bit 0 $\rightarrow |0\rangle$
 - Bit 1 $\rightarrow |1\rangle$
- X-basis (Diagonal Basis):
 - Bit 0 $\rightarrow (1/\sqrt{2}) (|0\rangle + |1\rangle) \equiv |+\rangle$
 - Bit 1 $\rightarrow (1/\sqrt{2}) (|0\rangle - |1\rangle) \equiv |-\rangle$

When a qubit is measured in a basis different from the one it was prepared in, the result is probabilistic. For instance, measuring $|+\rangle$ in the Z basis yields 0 or 1 with equal probability.

This probabilistic nature ensures that any eavesdropping attempt disturbs the system, making BB84 a secure foundation for quantum key distribution.

Implementation Overview

- 1) **Random Bit and Basis Generation:** The code starts by making random 0s and 1s (bits) for Alice to send. It also picks random “bases” (like settings on a filter) for both Alice and Bob. This randomness is important — it makes sure no one else can guess what’s happening.
- 2) **Quantum State Encoding and Measurement:** Based on Alice’s bit and basis, she prepares a qubit (a tiny piece of quantum data). Bob then measures that qubit using his own randomly chosen basis. If his basis matches Alice’s, he has a good chance of measuring the right bit. The code uses Qiskit to simulate this quantum process.
- 3) **Basis Matching (Sifting):** After all the qubits are sent and measured, Alice and Bob compare their basis choices (not the actual bits). They only keep the bits where their bases matched. This step is called sifting, and it's a normal part of the BB84 protocol.
- 4) **Iterative Key Accumulation:** One round usually isn’t enough to get a full key. So, the program repeats this whole process in small batches, adding more bits to the shared key each time. This loop continues until Alice and Bob have at least 128 matching bits to form a full key.
- 5) **Encryption and Decryption:** Once they have the shared key, Alice creates a random 128-bit message. She encrypts it by combining the message with the key using XOR (a simple logic operation). Bob uses the same key to decrypt it. If the original message and the decrypted message match, that means the key exchange worked perfectly.

Results

Starting BB84-style key exchange...

----- Iteration 1 -----

Matched indices: [0, 1, 7, 9, 10, 12]

Sender bits : [1, 1, 0, 1, 0, 0]

Receiver bits : [0, 1, 0, 1, 0, 1]

Accumulated key length: 4

----- Iteration 2 -----

Matched indices: [1, 3, 5, 7, 10, 12, 13, 14, 15]

Sender bits : [1, 1, 0, 0, 0, 1, 1, 1, 1]

Receiver bits : [0, 0, 1, 1, 1, 0, 1, 1, 1]

Accumulated key length: 7

----- Iteration 3 -----

Matched indices: [0, 1, 2, 5, 6, 7, 8, 10, 12, 13, 14]

Sender bits : [0, 1, 1, 0, 0, 1, 1, 1, 1, 0, 0]

Receiver bits : [0, 1, 0, 0, 1, 1, 0, 0, 0, 0, 1]

Accumulated key length: 12

----- Iteration 4 -----

Matched indices: [1, 2, 3, 6, 7, 11, 12, 13, 14, 15]

Sender bits : [1, 0, 0, 1, 1, 1, 0, 0, 0, 1]

Receiver bits : [0, 0, 0, 1, 0, 0, 1, 1, 0, 0]

Accumulated key length: 16

----- Iteration 5 -----

Matched indices: [1, 2, 5, 6, 7, 9, 11, 12, 15]

Sender bits : [1, 1, 0, 0, 1, 0, 0, 0, 0]

Receiver bits : [0, 0, 1, 0, 0, 0, 1, 0, 0]

Accumulated key length: 20

----- Iteration 6 -----

Matched indices: [3, 4, 5, 6, 7, 9, 11, 13]

Sender bits : [1, 1, 0, 0, 1, 1, 1, 1]

Receiver bits : [0, 1, 0, 0, 0, 1, 0, 0]

Accumulated key length: 24

----- Iteration 7 -----

Matched indices: [1, 2, 3, 6, 11, 14]

Sender bits : [0, 0, 0, 0, 1, 0]

Receiver bits : [0, 0, 0, 0, 0, 0]

Accumulated key length: 29

----- Iteration 8 -----

Matched indices: [0, 1, 2, 4, 6, 7, 8, 9, 10, 12, 14]

Sender bits : [0, 1, 1, 1, 1, 0, 0, 0, 1, 1, 0]

Receiver bits : [0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0]

Accumulated key length: 35

----- Iteration 9 -----

Matched indices: [0, 2, 4, 6, 7, 8, 9, 13, 15]

Sender bits : [0, 1, 1, 1, 1, 1, 1, 1, 1]

Receiver bits : [1, 1, 1, 0, 0, 0, 0, 0, 1]

Accumulated key length: 38

----- Iteration 10 -----

Matched indices: [0, 1, 2, 4, 6, 7, 9]

Sender bits : [0, 1, 0, 0, 1, 1, 1]

Receiver bits : [1, 0, 0, 0, 0, 0, 0]

Accumulated key length: 40

----- Iteration 11 -----

Matched indices: [0, 5, 7, 9, 12, 13, 14, 15]

Sender bits : [1, 1, 0, 1, 0, 0, 0, 1]

Receiver bits : [1, 0, 0, 1, 0, 0, 0, 0]

Accumulated key length: 46

----- Iteration 12 -----

Matched indices: [0, 1, 2, 4, 5, 6, 7, 11, 12, 15]

Sender bits : [1, 0, 1, 0, 0, 0, 0, 1, 1, 1]

Receiver bits : [0, 0, 0, 0, 0, 0, 1, 1, 0, 1]

Accumulated key length: 52

----- Iteration 13 -----

Matched indices: [0, 2, 3, 6, 7, 8, 10, 12, 13, 14, 15]

Sender bits : [0, 1, 0, 1, 1, 0, 1, 1, 0, 1, 1]

Receiver bits : [0, 0, 1, 0, 0, 0, 0, 0, 1, 1, 0]

Accumulated key length: 55

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----- Iteration 14 -----
Matched indices: [1, 3, 4, 5, 7, 8, 10, 11, 12, 15]
Sender bits      : [0, 0, 0, 1, 0, 0, 0, 1, 1, 0]
Receiver bits    : [0, 0, 0, 0, 0, 1, 0, 1, 0, 1]
Accumulated key length: 61

----- Iteration 15 -----
Matched indices: [0, 1, 2, 3, 5, 10, 11, 12]
Sender bits      : [1, 0, 0, 0, 1, 0, 1, 1]
Receiver bits    : [1, 0, 0, 0, 0, 1, 0, 0]
Accumulated key length: 65

----- Iteration 16 -----
Matched indices: [0, 1, 3, 4, 5, 11, 12, 13, 14, 15]
Sender bits      : [1, 0, 1, 1, 0, 0, 0, 0, 1, 0]
Receiver bits    : [1, 1, 1, 0, 0, 0, 1, 1, 0, 0]
Accumulated key length: 70

----- Iteration 17 -----
Matched indices: [1, 3, 5, 8, 11, 15]
Sender bits      : [1, 0, 0, 1, 1, 0]
Receiver bits    : [0, 0, 0, 0, 0, 0]
Accumulated key length: 73

----- Iteration 18 -----
Matched indices: [1, 2, 5, 6, 8, 11, 12]
Sender bits      : [0, 0, 1, 0, 0, 1, 0]
Receiver bits    : [0, 1, 0, 0, 1, 0, 0]
Accumulated key length: 76

----- Iteration 19 -----
Matched indices: [0, 3, 4, 5, 6, 10, 11, 15]
Sender bits      : [0, 0, 1, 1, 1, 1, 1, 1]
Receiver bits    : [1, 1, 0, 0, 0, 0, 0, 1]
Accumulated key length: 77

----- Iteration 20 -----
Matched indices: [1, 2, 4, 5, 7, 8, 10, 11, 12]
Sender bits      : [1, 1, 1, 1, 0, 0, 0, 1, 0]
Receiver bits    : [0, 0, 1, 0, 1, 0, 1, 1, 0]
Accumulated key length: 81
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----- Iteration 21 -----
Matched indices: [2, 6, 10, 11, 12, 13, 15]
Sender bits      : [0, 1, 1, 1, 0, 1, 1]
Receiver bits    : [0, 1, 1, 0, 0, 0, 0]
Accumulated key length: 85

----- Iteration 22 -----
Matched indices: [1, 3, 4, 5, 6, 7, 9, 13, 14, 15]
Sender bits      : [0, 1, 1, 0, 0, 1, 1, 0, 1, 1]
Receiver bits    : [1, 0, 1, 0, 0, 0, 1, 0, 1, 1]
Accumulated key length: 92

----- Iteration 23 -----
Matched indices: [1, 3, 4, 8, 9, 10, 12, 13, 14, 15]
Sender bits      : [1, 1, 1, 1, 1, 0, 0, 1, 0, 0]
Receiver bits    : [0, 0, 0, 0, 0, 1, 1, 1, 0, 0]
Accumulated key length: 95

----- Iteration 24 -----
Matched indices: [1, 5, 7, 8, 11, 12, 13, 15]
Sender bits      : [1, 0, 0, 0, 1, 1, 1, 0]
Receiver bits    : [0, 1, 1, 0, 0, 1, 1, 0]
Accumulated key length: 99

----- Iteration 25 -----
Matched indices: [0, 2, 6, 11, 15]
Sender bits      : [1, 1, 1, 1, 0]
Receiver bits    : [0, 0, 0, 0, 0]
Accumulated key length: 100

----- Iteration 26 -----
Matched indices: [1, 2, 4, 6, 11, 12, 14]
Sender bits      : [1, 0, 0, 0, 0, 0, 0]
Receiver bits    : [0, 0, 0, 0, 0, 0, 0]
Accumulated key length: 106

----- Iteration 27 -----
Matched indices: [1, 2, 5, 6, 8, 12, 13, 14, 15]
Sender bits      : [0, 0, 1, 0, 1, 1, 1, 0, 0]
Receiver bits    : [0, 1, 0, 0, 0, 1, 0, 0, 0]
Accumulated key length: 111
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----- Iteration 28 -----
Matched indices: [1, 5, 6, 7, 9, 13, 14]
Sender bits    : [1, 0, 1, 1, 0, 1, 1]
Receiver bits  : [0, 1, 0, 0, 0, 0, 1]
Accumulated key length: 113

----- Iteration 29 -----
Matched indices: [4, 6, 7, 9, 11]
Sender bits    : [1, 1, 1, 0, 1]
Receiver bits  : [1, 0, 1, 0, 0]
Accumulated key length: 116

----- Iteration 30 -----
Matched indices: [3, 5, 8, 12, 13, 15]
Sender bits    : [0, 0, 1, 0, 0, 0]
Receiver bits  : [0, 0, 0, 0, 0, 0]
Accumulated key length: 121

----- Iteration 31 -----
Matched indices: [0, 4, 5, 7, 8, 9, 10, 12, 14]
Sender bits    : [1, 1, 1, 0, 1, 1, 0, 0, 0]
Receiver bits  : [0, 0, 0, 1, 0, 0, 0, 0, 1]
Accumulated key length: 123

----- Iteration 32 -----
Matched indices: [0, 1, 2, 3, 6, 11, 13]
Sender bits    : [0, 1, 1, 1, 1, 1, 1]
Receiver bits  : [0, 0, 0, 0, 1, 0, 1]
Accumulated key length: 126

----- Iteration 33 -----
Matched indices: [0, 6, 7, 10, 11, 12, 13]
Sender bits    : [1, 0, 0, 0, 1, 0, 1]
Receiver bits  : [0, 0, 0, 0, 0, 1, 1]
Accumulated key length: 130

Shared key established successfully!
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This project presents a working simulation of the BB84 Quantum Key Distribution protocol using Qiskit. The code models quantum encoding, probabilistic measurement, and classical basis comparison through iterative communication between Alice and Bob. The protocol successfully demonstrates how a secure symmetric key can be established over a potentially insecure quantum channel. Through the final XOR-based encryption, the project confirms the effectiveness and correctness of the QKD process. This simulation lays the groundwork for understanding real-world quantum-secure communication systems.